



Gray Subcontractor Safety Orientation Manual 2023



Dear Subcontractor,

Safety is Gray's number one Core Value. "We put safety and quality of life first." Our goal is to ensure that all workers across our projects go home to their loved ones at the end of each day. We all play a key role in successfully achieving this goal. Putting safety and quality of life first ensures a safe, efficient, and productive worksite for us all. It is important for you to take the time to carefully read this Orientation Manual and discuss it with your employees. This manual will explain the obligations your Company has for safety while on our projects. Most of these obligations are clarifications of OSHA requirements, but there are some areas in which Gray exceeds OSHA requirements. Gray feels these policies make for the safest environment for all employees on our projects. Gray reserves the right to make modifications to this manual without prior notice, but Gray will use its best efforts to keep you apprised of modifications (or provide you with access to the current version of this manual).

Below are some of areas in which Gray's requirements exceed the requirements established by OSHA. Please note that this list is not all inclusive.

Required Gear:

- Safety glasses must be worn at all times.
- Safety helmets must be worn at all times.
- Helmet style safety helmets meeting both Z-89 and EN12492 worn with chin straps buckled.
- A 6-foot trigger height is required for all fall protection.
- Self-Retracting Lifelines (SRL) are required for fall protection in all aerial and scissor lifts.
- There is a mandatory glove use policy during all construction activities (ANSI Cut Level 4 and/or task specific).
- High-visibility shirts/vests must be worn by all parties on the jobsite.
- Safety toe work boots must be worn by all parties on the jobsite, this includes when wearing rubber boots.
- Metatarsal Guards must be worn when using any manually operated ground compaction equipment.

Prohibited Gear:

- Metal fish-tapes
- Stilts
- Aluminum ladders or manufactured wood ladders (Ladders should be used as a last resort on the project; Refer to section 33 of the manual for additional details)
- Headphone and earbud use is prohibited inside the construction site.
- The use of personal radios and/or speakers are prohibited from use on Gray sites.

Other Requirements:

- Safety Representatives and Designated Safety Representatives
- Gray Safety Analysis (GSA's)
- All excavations must be barricaded.
- Third party, post-assembly crane inspections are required on cranes that necessitate on site assembly.
- A 20-foot approach distance to over-head power lines is required.
- Hard Barricades around all temporary and/or permanent power panels, boxes, transformers, and or other electrical equipment in areas exposed to equipment traffic.
- A roof monitor is not allowed in lieu of fall protection.
- Spotters must be used when in congested areas, moving mobile equipment, working next to installed equipment and when any overhead work is being performed, if no barricade is in place (Refer to section 28 for additional information on Spotters).
- Overhead work must have an enclosed barricade or spotter in place prior to starting the work (Refer to section 28 for additional information on Spotters).
- Tool tethering when engineering and/or administrative controls cannot effectively prevent the hazard of dropped tools.
- Carbon Monoxide awareness (Refer to section 57 for additional information)
- Participation in the Mentoring Program for inexperienced workers with less than 1 year experience (See Section 52 for additional details).

There may be additional site-specific requirements that will be covered during site orientation.

If you are selected to work on this project, you and your Safety Representative will attend a Pre-Construction meeting to review the materials in this manual and discuss any questions your Company has.

Your Company has the non-delegable and sole responsibility for safety of your operations and for the safety of your employees and other persons engaged in the performance of your work. Your Company is required to train your employees on all safety requirements in this manual, as well as those required by OSHA, and federal, state, and local regulations, prior to starting work on the project. Your Company retains the responsibility to keep itself apprised of the current safety policies and procedures for the project.

Once on site, your employees will be required to watch a short safety orientation video explaining Gray's safety philosophy and general safety requirements for the project. Supervisors will be required to watch a separate video which sets forth expectations for all managers/supervisors on Gray worksites. At the end of the orientation process, your employees and supervisors will sign an Orientation Sign-off Sheet acknowledging the viewing of the safety orientation video(s).

It is important that we are proactive in the safety process and not reactive. We have invested in this orientation process to ensure Gray's safety expectations are communicated to all involved with our projects so that the project is completed in the safest way possible and so that all unsafe conditions are recognized and addressed.

Your Company is responsible for having your own written safety program that covers safety and health issues for your entire scope of work. For a safety program to be effective, it must be implemented and managed on site. We request your commitment to provide a safe workplace for all employees across the work site.

Thank you in advance for your commitment to safety. We look forward to working with you.

Sincerely,

Ned Brown, Director of Safety,
Gray Construction

Table of Contents

I. Significant Injury or Fatality (SIF)	9
II. Critical Life Saving Rules	9
III. Other Safety Violations	9
IV. General Guidelines	10
V. Contractor Site Supervision	10
VI. Subcontractor Safety and Health Policies.....	11
1. Alcohol and Drug Free Policy	11
2. Orientation	11
3. Workers with Disabilities	11
4. Fatigue Management.....	11
5. Project Safety Plan	11
6. Injuries/Incidents/Near Misses	12
7. Modified Duty Employee Program	12
8. First Aid/CPR/Bloodborne Pathogens Training.....	12
9. Subcontractor's Safety Representative/Designated Safety Representative/Safety Director.....	13
10. Competent Person Designation	14
11. Weekly Safety Meetings.....	14
12. Emergency Action Plan.....	14
13. Hazardous Communication Policies/Hazardous Materials	15
14. Dress Code.....	15
15. Fire Protection/Prevention	15
16. Lighting and Illumination.....	15
17. Confined Space Entry.....	16
18. Anti-Bias Policies	16
19. Work-Place Violence.....	17
20. Gray Safety Analysis (GSA)	17
21. Daily Job Site Inspection	17
22. Operator's Qualification Record.....	17
23. Phone Use	17
24. Hot Work Permits.....	17
25. Floor and Wall Openings	18
26. Fall Protection Policy	18
27. Aerial / Scissor Lifts	19
28. Spotters	19

29. Hand Tools	19
30. Head Protection	19
31. Tools & Equipment.....	20
32. Stairways	20
33. Ladders.....	20
34. Scaffolds.....	21
35. Respiratory Protection	21
36. Eye and Face Protection.....	21
37. Air Tools and Hoses.....	21
38. Silica	22
39. Welding and Compressed Gases	22
40. Excavation Protection Selection Form and Daily Inspection Form.....	22
VII. Gray's Ground Disturbance Form	23
VIII. Pot Holing	23
41. Control of Hazardous Energy/ Lockout Tagout Policy.....	23
42. Electrical Tools and Extension Cords	24
43. GFCI Program	24
44. Electrical	24
45. Cranes	25
46. Tilt-Up/Precast/Steel/Masonry Erection Bracing Requirements.....	26
47. Material Storage	26
48. Material Handling.....	27
49. Hand and Arm Protection	27
50. Hearing Protection	27
51. Working in Extreme Heat.....	27
52. Mentor Program.....	27
53. Stop the Drop.....	28
54. Lightning Policy.....	28
55. Pneumatic and Hydrostatic Testing of Piping Systems.....	28
56. Contractor's Responsibility for Site Safety.....	29
57. Carbon Monoxide	29
58. Health Crisis	29
59. Steel Decking.....	30
My Commitment to Safety Acknowledgment.....	31
Attachment #1: Work Procedures.....	32

1. Precast Tilt-Up Temporary Bracing Work Procedures	32
2. Steel Erection Temporary Bracing Work Procedure.....	33
3. Masonry Temporary Bracing Work Procedure	34
Attachment #2: Gray's Drug-Free Workplace Summary	36
1. Prohibited Conduct:	36
2. Kinds of Testing.....	36
3. Consequences	36
4. Inspections	36
5. Safety-Sensitive Positions	37
6. Confidentiality	37
7. Reservation of Rights	37
8. Random Testing	37
Attachment # 3: Americans with Disabilities Act	38

I. Significant Injury or Fatality (SIF)

Significant Injury or Fatality or "SIF" are incidents or potential incidents that have a significant or potentially significant impact on the safety and quality of life of the workers or visitors to our projects.

With the serious impacts associated with incidents in these areas, Gray will place significant emphasis on safety and focus on SIF potential. These areas of focus include, but are not limited to:

- Mobile equipment
- Working at heights
- Confined space
- Rigging and lifting operations
- Welding and cutting
- Lock out tag out and/or high energy potential
- Unplanned maintenance
- Unplanned changes in work
- Chemical handling

Gray has adopted a set of Critical Life Saving Rules as part of this planning effort.

II. Critical Life Saving Rules

The priority for every Gray project is the safety of every individual working on or visiting the project. As such failures in the following areas could have a significant impact on a person's safety and quality of life, they will be enforced as Critical Life Saving Rules.

Violations of the Critical Life Saving Rules will be enforced with written documentation and if determined in Gray's sole discretion, removal from the project. If violations in these areas occur, Gray's Corporate Safety department shall be notified, and a review meeting shall be held with the Contractor to determine if removal of the individual is permanent, or if a set of corrective actions can be established by the Contractor that will allow for the employee's return. The employee(s) shall not under any circumstance return to work until this review is conducted.

1. **Training** - Workers must be trained for their scope of work and safety requirements associated with it.
2. **Planning** - Proper planning of the work and the communication of the plan to the work force.
3. **LOTO** - Workers shall not work on live systems and LOTO procedures shall not be bypassed.
4. **Fall protection** - Fall protection shall be worn over 6 feet.
5. **Confined Space** - Do not go into a confined space without proper evaluation.
6. **Excavations** - Never enter an excavation that is not properly protected.
7. **Equipment** - Stay clear of mobile equipment, always get the operator's attention.
8. **Tools** - Provide the proper tools in a safe condition, all guards and safety devices shall be serviceable.
9. **Hazard Elimination** - First priority should be given to eliminating a hazard, not protecting against it.
10. **Leadership** - Managers must show strong safety leadership.

Additionally, any employee exhibiting a flagrant or willful disregard for safety, be it Gray, subcontractors or vendors, that could cause death, injury, or property damage, may be immediately removed from the project.

The Gray Site Team has the sole discretion to remove any employee on site if it is deemed necessary.

III. Other Safety Violations

All other Safety Violations observed shall be managed according to the procedures established below:

- First offense by an employee

Any team member observing a safety violation will verbally notify the employee's immediate supervisor of the violation and will request assistance. The employee will be given a verbal warning. The verbal warning will be logged with the Safety Representative.

- Second Offense (same employee)

Upon a second safety violation, a written report or warning will be filed with the onsite supervisor of the particular Subcontractor. The Gray Site Manager/Safety Representative will require a meeting with the employee and the supervisor. During this meeting, a Counseling Sheet will be completed and signed by everyone present.

- Third Offense (same employee)

Three safety violations by the same employee will result in his/her dismissal from the jobsite.

- Isolated Incident

The Gray Site Team has the sole discretion to remove any employee on site, if it is deemed necessary.

IV. General Guidelines

- All site employees shall use and wear appropriate fall protection when exposed to falls greater than 6 feet above a lower level and when working from aerial/scissor lift. Note: The “safety monitor” or “roof monitor” system for fall protection is not acceptable on Gray projects.
- Possession of or working under the influence of alcohol, illegal drugs or certain prescription drugs is prohibited.
- All temporary electrical equipment, including all connected by extension cords, shall be protected by Ground Fault Circuit Interrupters (GFCI).
- Be alert at all times and report all unsafe conditions or acts, along with all accidents and near misses to your supervisor immediately. A near miss is defined as, “an unplanned event that did not result in injury, illness, or damage, but if conditions were slightly different, had the potential to do so.”
- Proper work attire shall be worn at all times. This includes at a minimum 3-inch sleeves (note: some tasks require long sleeves), leather, safety-toe, above the ankle work boots, and long work pants (athletic style pants and sweatpants are not acceptable).
- Use of personal protective equipment such as safety helmets, safety glasses/goggles, high-viz shirts/vests, safety-toed work boots, and gloves are mandatory.
- “Horseplay” and other such unsafe acts are prohibited.
- Fire protection equipment is not to be tampered with or removed from its assigned location. Tampering with fire equipment is prohibited and may be grounds for termination.
- Prior to work beginning, all site employees are to visually inspect all hand tools and extension cords they will be using. Tools and extension cords that are found to be defective are to be tagged, taken out of service, and removed from the site or repaired immediately.
- Scaffolding shall be inspected for any defects by the subcontractor’s Competent Person prior to each work shift. If scaffolding is found to be incomplete or defective, it is to be immediately brought to the attention of the supervisor and tagged out of service until repairs are made or issues corrected and re-inspected by the Competent Person.
- Only authorized and properly instructed/trained/qualified site employees shall operate machinery equipment, vehicles, or tools.
- Vehicles are to be always operated in a safe manner by no one other than a trained and/or certified equipment operator.
- Utilize proper lifting techniques. Do not try to lift or push objects that are 50 lbs. or more. Get help if necessary.
- Do not enter barricaded areas unless authorized to do so.
- Modification or alteration of any piece of personal protective equipment is strictly prohibited.
- Defective or damaged personal protective equipment shall be removed from the site or repaired immediately.
- “No Smoking” rules shall be obeyed in posted areas.
- A formal, written plan for bracing will be required for the bracing of tilt-up and pre-cast panels, masonry walls, and steel erection activities. Further requirements of these plans may be found under the “Scope of Work” section of your contract. The steel bracing plan shall be reviewed and approved by the Gray Corporate Safety team prior to the start of work.
- The use of plastic fuel cans is prohibited.
- Contractors are required to perform housekeeping daily to keep all garbage and debris out of the work areas.
- Metal “fish-tapes” are prohibited on Gray sites.
- Mentoring Program guidelines will be followed.

V. Contractor Site Supervision

Site Supervision is considered the highest level of supervision for each Contractor on a Gray project. Each Contractor must assign a Site Superintendent or Site Manager to the project who will have overall leadership of, and who is accountable for, the safety performance of the Contractor’s workforce and tiered contractors on site. This individual must be a full-time employee and must possess the knowledge and leadership skills needed to complete the Contractor’s scope of work efficiently and effectively. Their focus should be on safety, quality, and schedule. Below is a list of some of the requirements for this role.

- Must attend the Pre-Construction meeting.
- Must attend Plan of the Day (POD) Meeting or Foreman meeting daily. The Foreman and essential leaders are also strongly encouraged to attend the POD or Foreman meeting daily.
- Must attend weekly coordination and/or any Scheduling/Safety meetings relating to the Project.

- Must have at least 5 years of experience in the scope of work being performed or must obtain approval from the Gray Project team based on experience, prior to the start of work.

VI. Subcontractor Safety and Health Policies

1. Alcohol and Drug Free Policy

Attached as Attachment #2 is a summary of Gray Construction's Drug-Free Workplace Policy & Procedures ("Gray's DFWPP"). Upon request, Gray will provide an electronic or hard copy of our full DFWPP. The Subcontractor acknowledges that they have reviewed Gray's DFWPP, and that the Subcontractor will comply with all requirements contained therein. The Subcontractor shall establish written procedures for compliance with Gray's DFWPP and agrees they will implement and enforce those procedures throughout the course of the project. Additionally, the Subcontractor agrees to follow all customer requirements, in addition to federal, state, and local laws and regulations regarding drug testing.

2. Orientation

All Subcontractors, tiered contractors, vendors, suppliers, and other personnel entering the project must complete the Gray Orientation process before they are allowed on site. All employees shall perform their orientation through the HammerTech system. The Subcontractor shall ensure all visitors and vendors sign in with the Gray Site team prior to accessing the site. The Subcontractor shall ensure that all their visitors, vendors, suppliers, and tiered contractors are supplied with all required Personal Protection Equipment ("PPE") prior to accessing site.

3. Workers with Disabilities

Construction projects keep communities moving forward. Still, construction can pose serious safety concerns and risks of injury to workers and pedestrians, and it can create unique safety risks and concerns for those with medical conditions. Gray and its subcontractors, tiered contractors, vendors, suppliers, and other personnel must remain vigilant and understand the need to identify and proactively address safety risks and concerns that are prevalent on construction projects, regardless of a worker's physical limitations. In doing so we can mitigate the risks of a construction-related accident or injury for all individuals on a construction site and provide reasonable accommodations to those with employees with disabilities as defined by the Americans with Disabilities Act (ADA). For additional information, please see Attachment #3.

4. Fatigue Management

Employee fatigue can present risks to safety, health, and productivity. If your workforce falls under any of the following categories, your Company is required to develop and implement a Fatigue Management Program. Your Fatigue Management Program should educate your workers on fatigue management techniques, set guidelines for management, and set controls to reduce the risks of incidents relating to worker fatigue from repetitive workdays with long hours without proper rest.

- Workers whose work hours exceed 10 hours a day for more than 5 days per week;
- Workers with work hours that exceed 60 hours in a 7-day work week;
- Workers whose work hours exceed 12 hours a day for more than 3 consecutive days; or
- Any Subcontractor that works a night shift.

5. Project Safety Plan

Gray's Project Management team will coordinate with the customer to establish any general site policies and procedures that are more stringent than those established by OSHA, Gray and/or the Subcontractor. These additional requirements shall be communicated to all Subcontractors on site prior to start of work. The Subcontractor shall create a Project Safety Plan and shall incorporate any state, local, or owner requirements that are more stringent than federal requirements or Gray's policy into the Project Safety Plan so it can serve as a site-specific document. It is the Subcontractor's responsibility to identify rules and regulations that are applicable to your Company and to train your employees accordingly. When developing the Project Safety Plan, particular focus should be given to areas with Significant Injury or Fatality ("SIF") potential. These areas include, but are not limited to:

- Mobile equipment
- Lock out and/or high energy potential
- Working at heights
- Unplanned maintenance
- Confined space
- Unplanned changes in work

- Rigging and lifting operations
- Chemical handling
- Welding and cutting

Customer specific requirements can include, but are not limited to:

- Customer specific orientation procedures;
- Prohibitions against cameras and other recording devices including cell phones;
- Permitting policies and/or procedures (Hot Work, Confined Space, Roof Access, etc.);
- Emergency evacuation policies and/or procedures;
- Incident notification protocols; and/or
- Special PPE requirements.

6. Injuries/Incidents/Near Misses

All incidents resulting in physical injuries or property damage, involving the workforce, public or visitors, and any Near Misses where there is the potential for injury and/or property damage, shall be reported to Gray's Site Management team at the time of occurrence by the contractor in charge of the worker(s)/work. A Near Miss is an unplanned event that did not result in injury, illness, or damage, but if conditions were slightly different, had the potential to do so.

The Gray Site Manager/Safety Manager shall ensure that all Subcontractors follow these reporting procedures.

- All injuries, property damages, and Near Misses must be reported, regardless of severity.
- Subcontractor shall report to Gray immediately, but in no event later than 2 hours after any incident that occurs at the site, or a shorter period, if required by the owner.
- In the event an employee must leave site for further medical evaluation, Gray should be notified prior to the injured employee leaving the site.
- Subcontractor shall provide Gray with a written incident report, including witness statements and photographs, within 8 hours of the incident, or sooner, if required by the owner.

7. Modified Duty Employee Program

Gray requires all Subcontractors performing work on any Gray construction site to participate in the Modified Duty Employee Program ("MDEP"). The purpose of a MDEP is to rehabilitate an employee with a work-related injury to full work capacity by allowing him/her to perform modified duties until he/she can return to work on a regular schedule with no restrictions.

The Gray Site Management team shall confer with any employee who suffers a work-related injury to discuss the type of injury and any prescribed limitations as documented by the treating physician. The goal of this conference is to ensure the employee can maintain

a regular on-site working schedule, but with duties which are modified to meet their limitations. Documentation of any limitation prescribed by the employee's treating physician should be submitted to the Gray Site Management team by the Subcontractor. The Subcontractor must also provide the Gray Site Management team with periodic updates on the employee's progress towards receiving a return-to-work release with no restrictions. It is the Subcontractor's responsibility to maintain an updated record of each modified duty employee with the type of injuries and limitations, as well as a record of scheduled physician visits and the employee's progress. Each time the employee has a return visit to the physician, the Gray Site Management team shall have a conference with the employee and Subcontractor's Representative to review the updated limitations and re-evaluate job duties.

8. First Aid/CPR/Bloodborne Pathogens Training

Each Subcontractor shall have at least one employee on site during all work activities who has been currently received CPR, First Aid and Bloodborne Pathogens ("BBP") training. There shall be an appropriate number of First Aid Responders on site for the size and scope of the work being performed. Subcontractor must also provide enough first aid kits and BBP kits for the size of Subcontractor's workforce. Copies of wallet cards or other acceptable means identifying CPR/First Aid Responders shall be furnished to Gray by Subcontractor and will be kept on file in Gray's site office. All workers trained in CPR, First Aid and BBP must be singularly identified by some obvious means.

In the event of an incident that could expose employees or equipment to potential exposure of bloodborne pathogens, **only those who have proper and up to date BBP training shall assist in the cleaning and disinfecting process.** These individuals should:

- Use the appropriate BBP kit and the PPE included in the kit.

- Use an EPA registered disinfectant suitable to treat contamination/spills and to disinfect surfaces or a 1:10 solution of bleach to water to disinfect all contaminated surfaces. Any chemicals should be used in compliance with OSHA's Hazard Communication Standard 29 CFR 1910.1200.
- Immediately clean and disinfect any visible surface contamination from blood, vomit, or other body fluids.
- Dispose of all materials used to clean according to the guidelines set forth by the OSHA requirements for the cleaning and decontamination of bloodborne pathogens pursuant to 29 CFR 1910.1030.

9. Subcontractor's Safety Representative/Designated Safety Representative/Safety Director

A. Safety Representative

Although everyone on a Gray jobsite is responsible for safety, each Subcontractor will appoint a Safety Representative that will support and coordinate compliance of all safety rules and regulations outlined in the Project Safety Plan (See Project Safety Plan for additional details).

The Safety Representative:

- Must hold an OSHA 30-hour card in construction issued within the past 10 years.
- May have other duties assigned to him/her in addition to his/her safety oversight and administration duties but if the Subcontractor is working 2 or more shifts, a separate Safety Representative shall be assigned to each shift and shall remain on site for the duration of work activities.
- Will attend and actively participate in Safety Committee meetings and other safety coordination and planning activities held by the Gray Site Management team.
- Will conduct a documented safety audit each shift and shall submit the audit to Gray (the Designated Safety Representative may also conduct these audits).

B. Designated Safety Representative

In certain instances, Gray may require Subcontractors to provide a Designated Safety Representative ("DSR") whose sole duties will be to enforce and administer the Project Safety Plan. Subcontractor's DSR shall conduct daily safety inspections and provide the Gray Site team with documentation of all items that are found to be non-compliant. Examples of this documentation include but are not limited to an email, pictures, and formal audit.

A DSR shall be required when:

1. These specific scopes of work will require a designated safety representative regardless of number of employees working on the project. A variance to this requirement may be requested through Gray Corporate Safety Department. Additional Safety Representatives would be required at the same criteria outlined in item 2 of this section.
 - Demolition
 - Steel Erection
 - Precast/Tilt-up Construction
 - Electrical
 - Mechanical
 - Refrigeration
 - Masonry
 - Roofing
 - Process Equipment Installation
2. If a contractor is not covered in item 1 of this section, a DSR is required with 20 or more employees on the same shift at the site. Subcontractors will be required to have an additional DSR for every 50 employees beyond the initial 20 (i.e., 70 employees require 2 DSRs, 120 employees require 3 DSRs).
3. A Subcontractor's safety performance is deemed unacceptable.
4. The Subcontractor is performing "high-risk" work activity as identified by the Gray Management team.
5. At the discretion of the Project Site Manager, the Gray Field Operations team, the Gray Corporate Safety team, or the Gray Field Safety team.

A copy of the DSR's resume shall be provided to the Gray Site Manager and the Gray Site Safety Representative at the Subcontractor's Pre-Construction meeting. Gray reserves the right to disqualify Subcontractor's SR's and/or DSR's based on the review of their resume or performance. Should Subcontractor fail, after 24 hours verbal notice from Gray, to provide an acceptable DSR as required by this section, Gray shall have the option to directly hire a DSR of Gray's choosing and charge all costs to the Subcontractor, as well as set-off all costs incurred against any monies due Subcontractor.

A prime contractor may provide safety representation (a 30-hour trained person "SR") for their tier but, the tier's manpower will be

added to that of the prime. Example: A steel erector has 15 employees on site, and they bring on a tier to install decking. The tier brings 10 people to the project and does not have their own Safety Representative. The steel worker can provide it, but then would have 25 people under his/her responsibility. The prime contractor would then have to provide a DSR, per Gray policy, because the prime has 20 or more employees on the same shift at the site (see Designated Safety Representative section above).

C. Subcontractor's Safety Director

Subcontractor's Safety Director, or an individual responsible for overall safety management for the project region, must regularly visit the project and provide documentation of the visit to the Gray Site Management team. The frequency of these visits will be established in the Pre-Construction meeting and will be based on safety performance on previous/other projects, scope, and schedule. These visits may also shift in frequency during the project based on performance.

10. Competent Person Designation

All Contractors and Subcontractors on the project will designate a "Competent Person(s)" as described by OSHA 29 CFR 1926 Safety Standards for the Construction Industry. A Competent Person is one who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authority to take prompt corrective measures to eliminate them.

A "Competent Person Acknowledgement Form" will be completed for each individual who is designated as Subcontractor's Competent Person. In addition, any documentation supporting the Competent Person designation shall be provided to the Gray Site Management team prior to the start of work on the project. If there is a change in Competent Person(s), this form must be updated and resubmitted to the Gray Site Management team before the new Competent Person accesses the jobsite.

The Competent Person Designation will include, but is not limited to the following areas:

- Fall Protection
- Ladders
- Confined Space Entry
- Scaffolding
- Tilt Panel Operations
- Trenching/Excavating
- Cranes/Rigging
- Electrical
- Steel Erection
- Respiratory Protection
- Silica exposure

11. Weekly Safety Meetings

Each Subcontractor shall have their appointed representative at any and all of Gray's Safety meetings and must have at least one weekly Safety meeting with their employees. A copy of the notes from the Subcontractor's employee Safety meeting must be submitted to the Gray Management team.

Each project shall conduct one site wide safety meeting each week (Minimum). This meeting must be attended by all personnel working on the project site on the day of the meeting. Attendant sign In sheet is required.

12. Emergency Action Plan

All of Subcontractors employees must be made aware of the elements included in Gray's Emergency Action Plan. An emergency is any situation that poses an immediate threat to life (workers, visitors, general public) or property. An emergency includes, but not be limited to:

- Fire
- Release of toxic gases
- Natural gas
- Explosion
- Dust
- Weather Emergency
- Flooding
- Fumes

- Smoke
- Equipment failure
- (i.e., collapse of a crane)
- Injuries and/or property damage

A Site-Specific Emergency Action Plan will be covered during Site Orientation. A written copy of the Site-Specific Emergency Action Plan will be posted in the Gray trailer and made available to all Subcontractor's on site. It is the Subcontractor's responsibility to ensure that their employees are aware of all policies and procedures within this plan. The Emergency Action Plan may include elements such as:

- Where to report to in case of a fire
- Where to take shelter in the case of a tornado warning
- How site workers will be made aware of these hazards
- Although Gray will develop a Site-Specific Emergency Action Plan, OSHA requires each employer to develop, maintain, and update their own Emergency Action Plan. If the plan developed by your Company deviates from the Gray Emergency Action Plan, you must notify the Gray Site team of any differences between your plan and the Gray Emergency Action Plan.

13. Hazardous Communication Policies/Hazardous Materials

A copy of the Subcontractor's Hazard Communication Program must be maintained on site along with all Safety Data Sheets ("SDS") for chemicals present on the project. The Subcontractor must train its employees on its policy. Subcontractor shall submit to the Gray Site Manager/Safety Representative all HAZCOM manuals, along with SDS for all chemicals only used on site with a site-specific index. A copy of all HAZCOM programs and Safety Data Sheets shall be centrally located in the on-site Gray Jobsite trailer. Hazardous materials must be transported in approved containers and be properly labeled.

Subcontractors using hazardous materials on site must alert other contractors of the potential hazards associated with the materials they are using. When working with or around hazardous materials, site employees must be aware of signs of overexposure, such as dizziness, nausea, trouble breathing, etc. If overexposure occurs, the employee should immediately remove himself/herself from the area and notify a supervisor.

14. Dress Code

It is the Subcontractor's responsibility to ensure that its employees, tier contractors, and others under the Subcontractor's supervision adhere to the following dress code:

- Shirts with 3-inch sleeves, some tasks may require long sleeves; long work pants (no athletic or sweatpants); and leather, safety-toe, above the ankle work boots are required.
- Clothes must be in good condition. Loose or ragged clothing, which may become a hazard around moving equipment or catch fire easily, are prohibited.
- Safety-toe work boots are to come above the ankle and be constructed of leather or other strong durable material and be in good condition. Canvas shoes, sandals, or open-toe shoes are prohibited. Waterproof rubber boots may be worn when conditions require them but still must be safety toe.
- High-visibility clothing is required at all Gray sites and must be worn on the outer-most layer. At a minimum, employees are required to wear hi-visibility shirts or vests. In certain circumstances a greater class of high-visibility clothing may be required, such as roadwork or per the owner's requirements.
- For work in non-daylight hours, high-visibility jackets or vests with reflective material that meet at a minimum, the ANSI Class II requirements must be worn.

15. Fire Protection/Prevention

Below is a listing of requirements relating to fire protection and prevention.

- Fire extinguishers will be located within 25 feet of all flammable/combustible liquids.
- Fire extinguishers shall be readily accessible to all work activities on the site.
- Any Subcontractor responsible for fire watch activities must ensure their employees are trained in fire extinguisher use.
- The Subcontractor is responsible for providing their own fire extinguishers where work activities warrant them.
- All mobile equipment shall be equipped with fire extinguishers.

16. Lighting and Illumination

The Subcontractor shall ensure that all temporary lights are equipped with guards to prevent accidental contact with the bulb. Guards are not required when the construction of the reflector is such that the bulb is deeply recessed. All temporary lighting

must be on a dedicated circuit.

17. Confined Space Entry

All Confined Space work shall be done within the scope of 1926.1200 (Subpart AA) and all permit required Confined Spaces shall be identified with a "Permit Required Confined Space" as they are constructed/arrive on site. Subcontractor shall provide Gray with a written copy of any and all information obtained by the Subcontractor regarding any Confined Space. These written materials will be available on site and made available to any and all contractors that may enter Confined Spaces.

A Confined Space is any space which is: 1) large enough and so configured that an individual can bodily enter and perform assigned work; 2) has limited or restricted means of entry or exit; and 3) is not designed for continuous employee occupancy.

Confined Spaces may include, but are not limited to:

- Boilers
- Vaults
- Cisterns
- Septic tanks
- Manholes
- Wells
- Furnaces
- Pumping stations
- Tank cars
- Tunnels
- Diked areas
- Storage bins
- Pits
- Sewers
- Silos
- Hoppers
- Vessels

Gray shall be notified prior to any Contractor allowing employees to perform work in a Confined Space. Employees shall not enter any Confined Space (i.e., manholes, equipment vaults, and underpinning operations) unless it has been tested for oxygen content, absence of flammable and/or toxic vapors, and all other potential hazards.

Hazards found in Confined Spaces may include but are not limited to:

- Oxygen deficient or oxygen enriched atmospheres
- Entrapment or configuration hazards
- Combustible/ flammable/ explosive gases and vapors
- Engulfment hazards
- Electrical
- Temperature extremes
- Toxic gases or vapors
- Mechanical hazards
- Combustible dust
- Corrosive chemicals

18. Anti-Bias Policies

It is the goal of Contractor to promote a work environment at the Project that is free from harassment of any kind. Contractor has ZERO TOLERANCE for harassment, including harassment on the basis of race, sex, gender, gender identity, gender expression, transgender status, sexual orientation, pregnancy, childbirth and other pregnancy-related conditions, color, national origin, ancestry, age, religious creed, citizenship, marital status (including registered domestic partners), parental status, physical disability, mental disability, medical condition, genetic information, military or veteran status (including protected veteran status), or any other characteristic or status protected by law.

Subcontractor agrees to be bound by the Policy Statement on Harassment and any violation or suspected violation of such policy

by Subcontractor or any of its officers, agents, servants, employees, subcontractors, or suppliers shall be considered as Subcontractor's failure to perform its obligations under the terms and conditions of this Agreement. Such failure shall be considered adequate and justifiable grounds for Contractor to effectuate its rights and remedies under provisions of this Agreement. Subcontractor shall actively promote a harassment-free work environment among its officers, agents, employees, subcontractors, and suppliers.

19. Work-Place Violence

Gray is committed to maintaining a workplace free from violent acts or threats of violence. If violent acts or threats or violence occur, Subcontractor agrees that it will work with Gray to appropriately respond to the situation. To this end, Gray seeks to provide a safe work environment to the full extent of the law. This statement shall not be construed to create an obligation on the part of Gray to take action beyond what is required by law. Subcontractors are required to comply with the requirements of the Gray Work-Place Violence policy contained in this Gray Safety Manual.

20. Gray Safety Analysis (GSA)

Subcontractor will be required to complete a Gray Safety Analysis (GSA) for each specific work activity. Each employee will sign and understand the GSA prior to the start of work. The GSA shall be available in the area of the activity for affected employees to view and sign. At the end of each day, or upon completion of the activity, employees are required to sign out of the task saying that they are leaving that task incident and injury free. The GSA will then be given to the Gray Management team.

Blank task specific GSAs shall be available in the Gray Jobsite trailer. Photo copying of a completed GSA is prohibited.

21. Daily Job Site Inspection

Each Subcontractor's Supervisor or Safety Representative is responsible for inspecting the work area. Any potentially unsafe issues identified by the Subcontractor must be brought to the attention of Gray's Site Management team prior to beginning any work and Subcontractor shall not begin work until identified issues have been resolved.

22. Operator's Qualification Record

Operators of heavy equipment shall be qualified. Equipment-specific documentation of each operator's qualification must be turned-in to the Gray Management team prior to beginning work. Subcontractors may be asked to turn in a duplicate key for equipment.

Operators shall inspect vehicles and equipment for defects at the beginning of each shift. Defective vehicles or equipment shall not be used. Drivers and operators of vehicles and equipment shall strictly observe speed limit restrictions and all posted traffic control signs.

Vehicles and equipment with a restricted view to the rear, and required to move in that direction, shall be equipped with an operable backup alarm, which is audible above surrounding work noise levels. In congested or critical areas, a spotter shall be required when the equipment is moving. If the situation dictates, multiple spotters shall be required. **(See section 28 for additional requirements on Spotters).**

Blades and buckets of earth-moving equipment shall be lowered to the ground when unattended and at the end of the workday. Engines shall be shut down during refueling operations. Running equipment shall not be left unattended.

Seat belts must be worn in all equipment/vehicles that are equipped with seat belts from the manufacturer.

23. Phone Use

The use of radios and earbuds are prohibited on Gray Projects while actively performing or in proximity of construction activities and maneuvering the project site. The use of cell phones, either work or personal, will be allowed on a limited basis but under no circumstances shall individuals be on their phone while performing a construction activity or while walking onsite. If the use of a phone is necessary, you must move to a safe area, clear of any work activity prior to utilizing the phone. For equipment operators, find a safe area to pull over, ensure the equipment brake is engaged and then use the phone.

The use of technology in the form of a tablet, I-pad or phone for inspection or auditing of the work is acceptable, but the user must establish themselves in a safe location.

Two-way radios may be used for communication purposes in equipment, however, if actively communicating, move to a safe area, clear of any work activity.

If there are any extenuating circumstances where these procedures cannot be followed, it must be communicated with and approved by Gray.

24. Hot Work Permits

A Hot Work permit shall be required for all spark-producing work activities taking place within an existing structure and/or when

required by the customer. Hot Work is any work that involves burning, welding, using fire- or spark-producing tools, or that produces a source of ignition.

When performing Hot Work, the following welding/cutting guidelines shall be followed:

- The work area shall be inspected for flammable solvents, vapors, and gases.
- Flammable and combustible materials shall be removed or covered.
- Appropriate fire extinguishing equipment shall be immediately available in the work area this shall include at a minimum a 10 lb extinguisher. Pending scope of hot work, additional or bigger fire extinguisher may be needed.
- A fire watch will be posted during welding/cutting operations for one hour following these operations when conditions make it appropriate. The fire watch will be instructed as per the requirements set forth in OSHA Regulation 1926.352(e) describing the duties of a fire watch.
- Fire watch required to stay in the area at least 60 minutes after completion of hot work. Customer requirements may exceed 60 minutes.
- When the building is 75% completed/enclosed a Hot Work Permit is required when Hot Work is taking place inside or within 25 feet of building

25. Floor and Wall Openings

When employees are exposed to overhead hazards, Subcontractors shall ensure the area is properly barricaded to restrict traffic in and out of the area. Toe boards, screens, or guardrail systems shall be erected to prevent objects and materials from falling onto lower levels. A "hole" is a gap or void 2 inches or more in its least dimension, in a floor, roof, or other walking/working surface. All covers shall meet the elements set forth in OSHA Regulation 1926.502(i).

An "opening" is a gap or void 30 inches or more high and 18 inches or more wide in a wall, partition, vertical walking working surface, or similar surface through which employees can fall to a lower level. All wall openings shall be guarded as set forth in 1926.501. Self-Retracting Lifelines (SRL) shall be required for fall protection in all aerial and scissor lifts (See Fall Protection Policy and Aerial / Scissor Lifts sections for additional details).

26. Fall Protection Policy

Gray has developed and implemented the following requirements regarding Fall Protection. Subcontractors shall ensure 100% COMPLIANCE with ALL FALL PROTECTION Regulations. Gray enforces a 6-foot trigger height for all work. Subcontractors are responsible for ensuring their employees are made aware of and for enforcing these policies. There will be Zero Tolerance for violations of Gray's Fall Protection Policies. Subcontractors found in violation of Fall Protection Policies may be removed from the site.

A. Fall Protection

- 100% Compliance with fall protection rules is required when employees are working at heights in excess of 6 feet and/or when working from aerial/scissor lifts.
- Gray has a ZERO TOLERANCE for Fall Protection Violations. Subcontractors found IN VIOLATION OF THIS POLICY MAY BE REMOVED FROM THE SITE AND/OR PROJECT, at Gray's discretion.
- All fall protection components are to be inspected daily and prior to every use. DO NOT USE DAMAGED or WORN-OUT COMPONENTS.
- Anchorage points must be rated for 5,000 LBS per person.
- All fall protection equipment shall be used properly based on OSHA and manufacturer guidelines.
- When a Subcontractor believes their work falls outside of standard fall protection practices, the Subcontractor may submit a written alternate fall protection plan to the Gray Safety team for review. No work that deviates from standard fall protection practices may be conducted until a meeting has been held with Gray Safety staff.
- A leading edge rated retractable must be used in when employees are in leading edge situations. This may include but not limited to installing metal decking, installing standing seam roofs, over sharp edges, etc.

B. Warning Line(s)

Warning lines shall be established and maintained by the Subcontractor(s) conducting work when there is a potential to fall to a lower level. Roofers shall maintain a warning line at 6 feet unless mechanical equipment is present, then that line shall be extended to 10 feet. A warning line shall be established at 15 feet for non-roofers that may be present on the roof.

C. Roof and Decking

Any employees who are working on decking or on the roof must attend a roof access training presentation. A permit must be completed for all decking and roof work to address and/or eliminate hazards. The permit is provided by the Gray Site Management team.

27. Aerial / Scissor Lifts

Subcontractors shall perform and document daily/pre-shift inspections of all lifts. Only authorized and professionally trained personnel may operate aerial and scissor lifts. Subcontractor shall adhere to the following.

- Lifts shall be used on surfaces that will fully support them.
- Lifts shall only be operated on level surfaces.
- Dead-man type safety switches and controls shall not be bypassed.
- Lifts shall be kept at least 20 feet from energized, overhead power-distribution lines.
- When lifts may be exposed to other electrical hazards such as cable trays, electrical service, switchboards, panelboards, circuit panels, the K tables in Subpart K of the 1926 standards must be followed. At no time may any lift be closer than 3 feet to any energized piece of equipment.
- Fall protection, including but not limited to, Self-Retracting Lifelines (SRL), are required in all scissor and aerial lifts. Employees using lifts must wear approved fall protection equipment and attach the lanyard to the engineered anchorage point at all times.
- Employees shall perform all work from lifts with both feet on the floor of the basket. No employees shall be permitted to perform work from the toe board, mid-rail, or top rail of a lift.
- Onsite modifications to lifts are prohibited.
- All attachments must be engineered and designed for the type of lift it is to be attached to.
- An Aerial Lift GSA shall be completed daily by each operator prior to operation
- Barricade and/or Spotters shall be used in the area where overhead work is taking place.
- All manufacturers' requirements shall be followed.
- Aerial lifts shall not be operated when high winds could affect the equipment's safe operation. Please refer to the manufacture's guidelines for further details.
- Tool tethering may be required based on where the work is taking place.

28. Spotters

A Spotter is necessary when the driver or operator of equipment or a vehicle does not have a full view of the travel path or when a piece of equipment is carrying materials in or out of a building. Two spotters shall be required when equipment is traveling in and out of doorways and/or entry ways.

A Spotter is also required in congested areas and when within 10 feet of installed owner's/customer's equipment, specialized equipment, installed equipment (ground level), in close proximity to structures, or when deemed necessary by the Gray Site team.

The Spotter(s) shall be competent and trained on the job site. The Spotter(s) will not have any additional duties while they are acting in that capacity and will not use personal mobile phones, headphones, or other items that can cause a distraction while performing their duties.

The Spotter(s) and operator shall have a pre-movement review of the line of travel. The discussion will allow all parties to identify potential hazards along the way. The Spotter and driver will always maintain visual contact while the vehicle is moving forward or backward. The operator will stop immediately if he/she loses sight of the Spotter(s).

Spotters shall remain visible to all other trade traffic during duties. If not, a second spotter shall be in place to divert other trade foot/ equipment traffic.

If necessary, the Spotter may be required to wear a different colored safety helmet or vest to be identifiable to operator/jobsite employees.

29. Hand Tools

All hand tools shall be inspected to ensure they are in good condition prior to the start of work. Wooden handles on tools shall be free of cracks or other defects. Tape or other coverings shall not be placed on handles. Impact surfaces of tools shall be free of mushroomed surfaces. Wrenches with sprung or worn jaws shall not be used. All defective tools must be immediately removed from the site.

30. Head Protection

Safety Helmets must be worn at all times by personnel on Gray sites and shall meet the specifications contained in American National Standards Institute, Z89.1-2014 and EN12492 standard, Safety Requirements for Industrial Head Protection. Below is a listing of additional requirements.

- Safety helmets for the protection of employees exposed to high voltage electrical shock and burns shall meet the specifications

of the American National Standards Institute.

- Helmet style safety helmets meeting both Z-89.1-2014 and EN12492 are required to be worn with chin straps buckled by everyone on the jobsite.
- Metal safety helmets shall not be worn.
- Safety helmets should be worn with the visor (bill) facing forward.
- When additional face protection is needed, it must be worn in conjunction with the safety helmet.

31. Tools & Equipment

Employees shall inspect all tools and equipment for defects prior to the start of work. Defective tools and equipment shall be tagged as defective and removed from the job site until repaired or replaced. Guards and safety devices shall be left in place and in operable condition. Any tools or equipment with missing or defective guards are prohibited. The Subcontractor is responsible for any and all tools brought on site by their employees, tiers, and others under their supervisor.

32. Stairways

Foot traffic is prohibited on stairways with pan stairs (except during stairway construction) unless the pans are filled with wood or other solid material to at least the top of the pan. Temporary stairways will be kept clear of debris and allow for easy egress.

A stair tower shall be provided, at a minimum, when:

- The building is 20 feet or higher in height
- The roof area of the building is 20,000 square feet or more.
- If you have multiple trades accessing the roof in addition to the decking sub or roofer.
- Toe boards or barricades must be provided, at a minimum. Each stair tower is required to be tagged for daily inspection.
- An adequate number of stair towers shall be erected for proper access and egress of the roof or elevated area.

33. Ladders

A. Ladders Last

This policy is implemented to ensure that Contractors are exhausting all available resources to avoid the use of step ladders, extension ladders, and job-made ladders to access elevated work areas. Examples of alternative means of access include but are not limited to aerial/scissor lifts, scaffolds, and stairways.

If it is decided that a ladder is the only viable option, the Ladders Last section of the GSA shall be completed prior to the work beginning. This policy will not affect the access/egress for excavations. In addition,

- Ladders shall be inspected prior to use. Defective ladders shall be tagged as defective and removed from the jobsite.
- Manufactured wood ladders, aluminum step ladders, and aluminum extension ladders are prohibited.
- Ladders shall be placed on a sound footing and shall not be placed on unstable objects such as loose bricks.
- Employees shall not carry tools, materials, or objects while climbing ladders.
- A-frame ladders shall be secured from displacement if side loading can occur, or the employee cannot maintain three points of contact.
- Extension ladders must be placed at an angle not to exceed one foot of run for every four feet of rise.
- Any employee working from a ladder must keep the core of their body inside the frame of the ladder. There may be specific work scenarios where employees will be required to wear personal fall protection systems while working from a ladder.
- Job-made wooden ladders are allowed when they meet the requirements set forth in CFR 1926.1053. However, the side rail shall extend 42 inches above the landing when used for access/egress.
- All ladders must have legible stickers with manufacturer's ratings and instructions.
- Ladders that are not in use should be removed from the vertical position.
- Ladders used to access mezzanine or similar elevated location shall have a walk-through opening with a safety chain or gate, at no time shall a person be required to fit through a handrail or cable rail to access ladder or landing.

B. Extension Ladders

Extension ladders shall be secured at the top and must extend 42 inches above the landing when used for access/egress. When used as semi-permanent access, the top and bottom of an extension ladder must be secured.

C. Step Ladders

Platform Ladders should be considered as the primary option when a step ladder must be used. Step ladders shall only be used in the open and locked position. Employees shall not work from the top cap or top rung of step ladders.

34. Scaffolds

Scaffold systems shall be erected and maintained based on the regulations set forth in 1926 Subpart L, as well as manufacturer's guidelines. Contractors shall provide a Competent Person to perform daily/pre-shift inspections whenever scaffolds are being utilized. Only competent scaffold builders are to erect or modify scaffolding. Employees responsible for the erection or dismantling of scaffolds shall be protected from falls 6 feet or greater by either guardrails or personal fall arrest systems when feasible. Any employee working on a scaffold 6 feet or greater shall be protected against falls by guardrails or a personal fall arrest system. Baker/Perry scaffold shall have guardrails attached when pic boards are set at 6 feet or greater and shall follow all manufacturers' requirements. "Hooches" shall not be attached directly to Scaffold components.

35. Respiratory Protection

Employers requiring their employees to wear respirators, including dust masks, shall submit a copy of their written Respiratory Program, as well as documentation naming all employees qualified to wear a respirator based on a medical evaluation and fit test. Further documentation may be requested (Refer to OSHA's Appendix D).

36. Eye and Face Protection

Safety glasses shall be worn 100% of the time on all Gray sites. When a building is enclosed, 75% clear glasses or indoor/outdoor glasses are to be worn. Dark glasses are to be worn only when outside. Eye and face protection equipment shall meet the requirements specified in American National Standards Institute, Z87.1-2015.

For some tasks, safety glasses do not provide adequate protection. Subcontractor shall evaluate potential hazards associated with specific tasks and establish minimum acceptable protection for those tasks. Some examples of these tasks include, but are not limited to:

- Insulation work
- Powder-actuated tools
- Chipping
- Sandblasting
- Impact wrenches
- Concrete breaking
- Catalyst handling
- Handling chemicals
- Burning
- Some woodworking tools
- Grinding
- Cutting
- Battery charging and jump starting
- Opening process lines or systems
- High pressure air or water blasting or cleaning
- Welding

Some examples of additional protection are:

- mono goggles
- burning and welding goggles
- chemical splash goggles
- welding hoods with appropriate filters
- face shields
- sandblasting hoods

37. Air Tools and Hoses

Attachments shall be secured to tools by proper retainers. Tools shall not be hoisted or carried by their hoses. Crows-foot type

connections on air hoses shall be safety-wired at each connection. Compressed air lines that are in use shall have a whip check at each connection.

38. Silica

Exposure to airborne silica occurs when cutting, sawing, drilling, and crushing concrete, brick, ceramic tile, rock, and stone products. Any Subcontractor with a potential to disturb silica must submit a written Silica Exposure Control Plan to Gray for their review prior to beginning any work that would disturb silica containing materials. The plan shall include documentation of training of employees on the Site-Specific Silica Plan. All plans must be in compliance with 1926.1153 (refer to Table 1 of OSHA's Silica standard). If Subcontractor is not following Table 1, proper testing and/or historical data must be submitted. If respirators are required by either the employer or any OSHA standard, proper documentation (health evaluation, fit test, etc.) must be provided (See Respiratory Protection section).

39. Welding and Compressed Gases

Please reference the following guidelines when your work involves welding or the use of compressed gases.

- Welding equipment shall be kept free of oil, grease, or other oil-based products.
- Screens shall be used when weld stations or fabrication areas are established.
- Compressed gas cylinders shall not be exposed to sparks, slag, flame, or other heat sources.
- Compressed air lines that are in use shall have a whip check at each connection.
- Flashback arresters located adjacent to the regulators are required on both the fuel and oxygen systems.
- Cylinders shall be secured in an upright position during use, storage, and transportation.
- Protective caps shall be in place over valve stems when regulators are not in place, when cylinders are in storage, and when cylinders are being transported or hoisted by motorized vehicles or equipment.
- Oxygen cylinders in storage shall be separated from fuel-gas cylinders or combustible materials, a minimum distance of 20 feet or by a noncombustible barrier at least 5 feet high having a fire-resistance rating of at least one-half hour.
- Cylinders shall not be taken into Confined Spaces.
- All cylinders must be identified with the contractor's name.
- Subcontractors shall follow Subpart Z regarding Toxic and Hazardous Substances.

40. Excavation Protection Selection Form and Daily Inspection Form

All Contractors performing Excavating/Trenching activities on site shall provide a "Competent Person" as defined by OSHA standards. The Competent Person shall be responsible for ensuring that employees are protected whenever working inside an excavation or trench. Subcontractors shall be required to select appropriate protection from cave-ins, in accordance with the OSHA Standards.

Subcontractors shall also ensure that excavations and trenches are inspected daily by the Designated Competent Person, and whenever conditions are present which may have altered, or changed the configuration of the trench/excavation, such as heavy rains, snow, etc. Inspection records shall be submitted to Gray daily. All excavations deeper than 5 feet must be sloped, benched, or shored. All excavations shall be inspected by the Designated Competent Person prior to employees entering the excavation and an assessment must be made as to whether protective systems are necessary.

A. Excavations

All excavations, regardless of depth, that do not have a hard guard, must be identified with an elevated, visual barrier. This barrier must be so designed as to prevent non-authorized personnel from entering the area. Some examples of barrier include but are not limited to:

- cones
- barrels
- snow fence
- danger/caution tape

Any excavation deeper than 6 feet that is not sloped or benched must be guarded to prevent falls. This guarding may be a warning line system 6 feet back from the edge of the excavation or a hard-guard system at the edge of the excavation. Any excavation deeper than 6 feet on site shall be evaluated by Gray as well as the Subcontractor's Competent Person to ensure employee and equipment safety.

In addition:

- Any excavation/trench 19 inches or greater in depth shall have an appropriate means of access/egress.
- If using a trench box, the top of the trench box should extend 18 inches above grade when feasible.

- Spoil dirt and equipment shall be kept at least 2 feet from the edge of the excavations.
- A stairway, ladder, ramp, or other safe means of egress shall be located in trench excavations that are 4 feet or more in depth to require no more than 25 feet of lateral travel for employees.

B. Underground Utilities

All underground utilities in the area must be located prior to excavating. This may consist of multiple steps including, but not limited to:

- Calling local “before you dig” services;
- Having a ground penetrating radar service conduct a survey;
- Obtaining as-built drawings from any previous work; and/or
- Working with contractors/owners who have completed prior work in area.

VII. Gray’s Ground Disturbance Form

A Gray Ground Disturbance form must be completed, and all parties listed on the form, or a representative, must complete their applicable portions prior to any excavating or ground disturbance.

VIII. Pot Holing

Once utilities are located and are known or suspected to be in the immediate excavation area, they must be visually located by a non-destructive means of excavating either by hand digging or hydro/air vacuum excavating. This must be done prior to excavating within 24 inches of the known utility.

A. Underground Utility Protection

All underground utilities and piping installed by a Gray Subcontractor shall be marked with locating material such as mag tape or tracer wire. The locating material shall be placed at a minimum of 24 inches above the underground utility unless otherwise specified by an engineer, local jurisdiction, or by the Gray Site Management team. The Gray Site Management and Gray Corporate Safety teams shall review and approve any requests for a variance or deviation from this requirement. Third party locating services may be required to find the location of utilities, including their depth.

Red Line drawings shall be updated daily to ensure proper location of installed underground utilities/piping. In addition, Red Line drawings/Ground Disturbance drawings shall be reviewed daily at the Ground Disturbance meeting.

B. Ground Disturbance Form Sign-Off Procedure

It is the Subcontractor’s responsibility to complete the Ground Disturbance Form for work to be performed, paying special attention to provide as much detail as possible to the location of the work. The Subcontractor will be present at the Ground Disturbance meeting held every morning. All MEP contractors will be required to attend this meeting daily. If Subcontractor fails to attend the Ground Disturbance meeting, NO digging will take place for that day.

The Subcontractor will mark up the drawing and review the previous day’s work on the overall site drawing. Each Subcontractor will have a specific color assigned to them for site drawing. Discussions will be made about the present days’ work. If lines are crossed on the site drawing, the dig area will be walked by the Subcontractors that are involved.

Subcontractors must make sure that red lines are being updated to ensure accurate drawings. These markups shall include exact measurements and locations of lines, including depth. Any digging within 2 feet of any type of existing line must be hand-dug.

Although these steps shall be followed, third party line locate, contractor locate, radar, and potholing shall still be performed.

41. Control of Hazardous Energy/ Lockout Tagout Policy

It is Gray’s policy that no “live” work is done on Gray sites. If a Subcontractor deems this policy infeasible, the Gray Safety team must be contacted prior to work that deviates from this policy. The Gray Site Management team shall conduct a Pre-Construction meeting with all Subcontractors likely to perform Lockout/Tagout (“LOTO”) on site.

The Pre-Construction meeting shall be held in the Gray jobsite trailer and attended by the following:

- Gray Site Management Team;
- Customer Maintenance Representative (if applicable);
- Customer Safety Representative (if applicable);
- Subcontractor Project Manager;
- Subcontractor Site Superintendent; and
- Subcontractor Safety Representative.

The Gray Site Management team shall ensure that all items established in the Electrical/Mechanical Pre-Construction meeting

are covered in detail at this meeting.

All circuits, systems, machinery, or equipment that can be locked out, shall be locked out prior to the start of work. Any employee required to work on these systems shall ensure a "Zero Energy State" is reached before work is performed.

Subcontractors performing work requiring LOTO shall develop specific procedures for control of hazardous energy sources (in addition to these general procedures) for any circuits, systems, machines, or equipment before any installation, maintenance, or servicing work is performed. All Lock-out/Tag-out procedures shall conform to all applicable OSHA standards and manufacturer's guidelines.

Zip ties/Tags shall not be used in lieu of locks.

42. Electrical Tools and Extension Cords

These guidelines should be followed when using electrical tools and extension cords:

- Extension cords used with portable electrical tools and appliances shall be of the three-wire type.
- Cords with the ground prong removed or rendered ineffective shall be removed from the jobsite.
- Cords shall be rated for hard or extra-hard usage.
- Only cords 12 gauge or better may be used on site.
- Electrical cords shall be covered and/or elevated to keep cords from creating a tripping hazard to employees or other persons in the area or causing damage to the cords.
- Extension cords shall be free of repair or splice, kept clear of traffic aisles, and not subjected to vehicular traffic.
- Ground Fault Circuit Interrupters (GFCI) shall be tested before use.
- Employees shall not operate electrical tools while standing in water.
- Electrical tools shall not be hoisted or carried by their power cords.

A monthly color code inspection is required for all electrical cords and corded tools. For extension cords, both ends of the cord shall be marked. The color codes for each month are as follows:

- | | |
|---------------------------|-------------------------|
| • January - <i>Red</i> | July - <i>Red</i> |
| • February - <i>White</i> | August - <i>White</i> |
| • March - <i>Blue</i> | September - <i>Blue</i> |
| • April - <i>Red</i> | October - <i>Red</i> |
| • May - <i>White</i> | November - <i>White</i> |
| • June - <i>Blue</i> | December - <i>Blue</i> |

Although this policy covers only electrical cords and corded tools, all other equipment on site that requires inspection (per the 1926 OSHA standards) must be conducted as the standard and/or manufacturer requires. This includes but is not limited to:

- Rigging
- Ladders
- Fall protection
- Fire extinguishers

43. GFCI Program

The use of Ground Fault Circuit Interrupters (GFCIs) is required at all times. The GFCI may be located at the service, the outlet, or a GFCI whip may be used. No circuit breaker will take the place of a GFCI on any power source. This requirement covers all voltages where a GFCI is commercially available.

44. Electrical

All temporary and permanent electrical work installation and wire capacities shall conform to the National Electrical Code and other applicable federal, state, and local codes. Only qualified electricians familiar with code requirements shall be allowed to perform electrical work. No employee shall be permitted to work on any energized electrical power circuit without first informing Gray Safety staff. In addition:

- All temporary electric wiring shall be installed so the wiring cannot be damaged.
- All temporary electric wiring shall be installed with MC cable. The use of Romex or like cabling is prohibited on Gray Projects.
- Temporary Panels shall be provided, and each service feeder or branch circuit at the point shall be legibly marked to indicate its purpose. On circuits exceeding 600 volts, "Danger - High Voltage" signs shall be posted where unauthorized persons might come in contact with live parts.

- Entrances to rooms and other locations containing exposed live parts shall be marked with conspicuous warning signs forbidding unqualified person to enter.
- Metal "Fish-Tapes" are prohibited for all scopes of work on Gray sites.
- Safe working distances should conform with 1926.403(i)(1)(i).
- Any Trailer Mounted Generators or Generator providing 480v power must be inspected by the Gray Site Management team prior to use on the project.

45. Cranes

All Crane Operators shall be NCCCO, NCCER, or OECP certified. A CIC Certification is acceptable if certification was obtained prior to 12/2/2019. Certified Crane Operators shall provide a copy of the written evaluation that complies with 1926 Subpart CC, App C. All Signalmen shall be "qualified" as defined by 1926 Subpart CC. The Subcontractor will be required to submit a copy of their Rigger/ Signal Person Qualification Program. The Subcontractor shall also submit a copy of the Lead Rigger's resume to Gray prior to a Critical Lift being performed.

The Ground Conditions Assessment form will be completed for cranes before work begins

A third-party crane inspection will be conducted after every assembly. Documentation of annual inspections, as well as monthly inspections, shall be submitted to Gray prior to work beginning.

For cranes that do not require "assembly," a qualified person must conduct a post set-up inspection. This inspection will be documented, and a copy of this documentation shall be submitted to Gray prior to the beginning of work. The Crane Operator may not be the qualified person for the post set up inspection.

Additionally:

- Operators shall not hoist improperly rigged or over capacity loads.
- Accessible areas of crane's rotating superstructure and counterweight shall be barricaded.

A. Inspections

The following inspections are required of all cranes and hoisting equipment:

- Shift inspections
- Monthly inspections
- Annual inspections
- Shift, monthly, and annual wire rope inspections

The top wind speed under which the crane can be safely operated based on the crane's current configuration shall be noted on the daily crane inspection form. Manufacturer guidelines shall be followed.

The following additional special inspections are required in particular circumstances:

- Post-assembly, third-party inspections
- Pre- and post-erection inspections of tower cranes
- Equipment used in severe service
- Inspections of certain modified equipment
- Inspections of certain repaired/adjusted equipment

All cranes shall be inspected in accordance with OSHA 29 CFR 1926.1412 requirements.

For those cranes that are required to be "assembled", the Subcontractor shall identify an Assembly/ Disassembly Director (A/D Director). This identification shall be made in writing to the Gray Site Management team prior to the start of work. The A/D Director shall meet the criteria for both a competent and qualified person. The A/D Director shall ensure the assembly/disassembly crew is familiar with their tasks, the hazards associated with the work, and hazardous positions/locations that should be avoided.

B. Rigging

Rigging may only be performed by a "qualified" or "certified" rigger.

Rigging equipment shall be inspected prior to each use. Defective equipment shall be tagged defective and removed from the site.

Loads must not be hoisted unless the weight is known, and the rigging equipment is adequate and properly attached. Capacities of rigging equipment shall not be exceeded. Engineered softeners shall be used on all edges when synthetic rigging is used.

Steel members (Joist, Girder, Beam etc.) must be mechanically secured on both ends, either bolted or welded, prior to releasing rigging.

C. Critical Lifts

A Critical Lift plan and GSA shall be submitted to the Gray Site Management team prior to work beginning for all lifts that include any of the following:

- When the operator may lose line of sight with the load (blind hoisting)
- When the lift includes more than one piece of equipment
- When the lift exceeds 75% of the cranes lifting capacity
- When the lift involves specialized equipment
- When hoisting personnel
- When using helicopter picks
- When the lift involves traveling with a suspended load from a crane
- As deemed necessary by the Gray Site Management team

46. Tilt-Up/Precast/Steel/Masonry Erection Bracing Requirements

Any prescriptions in this section are only project minimum standards. The Subcontractor remains solely responsible for planning and implementation of the erection bracing to safely support their erected elements.

A. Tilt-Up/Precast

Only employees critical to the wall operation are allowed in the exclusion zone. If non-essential employees must gain access to the exclusion zone, an Exclusion Zone Permit must be issued by the Gray Site staff.

The exclusion zone must:

- Be constructed of at least snow fence (safety fence).
- Be established at a distance of wall height plus 6 feet away from the wall(s) on the brace, as well as the non-brace side.

When set panels are at least 100 feet from a panel being erected, the exclusion zone on the brace side may be moved within 6 feet of the foot of the braces. When the exclusion zone can be moved to within 6 feet of the braces, pendant flagging may be used in lieu of snow fence. The non-brace side must still be maintained at wall height plus 6 feet with snow fence.

A copy of the bracing plan needs to be submitted to the Corporate Field Operations team prior to work beginning. Any modifications must be approved by a registered professional engineer and changes must be updated in the submittals provided to Gray. Any deviations or changes to a temporary bracing plan after the approval process must be resubmitted to Gray for review.

If "leaner" panels are required, the Subcontractor shall include this in their bracing plan. This should also include, location, all details of necessary components such as braces, tilt wall anchors, inserts, and bolts used for bracing. This must be approved and stamped by a Registered Professional Engineer. The bracing plan must include sequencing of erection.

No bracing may be removed until permission is given to Gray from a Registered Professional Engineer.

The bracing requirements in Attachment #1 apply to tilt-up panels, precast panels, as well as masonry walls and temporary steel bracing.

B. Steel

The erection shall begin at a braced bay. A braced bay is a bay braced with erection bracing on all four sides. As the erection proceeds beyond the braced bay, each subsequently erected column line must be braced with at least one erection brace. By way of illustration and not as a prescriptive drawing or specification, see Figure 7.1 from the AISC Design Guide 10 Erection Bracing of Low-Rise Structural Steel Buildings (Attachment 1).

A copy of Bracing Plan needs to be submitted to the Corporate Field Operations team prior to work beginning. Any deviations or changes to a temporary Bracing Plan after the approval process must be resubmitted to Gray for review. The Bracing Plan must include sequencing of erection, material being used as temporary bracing, attachment method, and must be signed by a Principle of the Company. No bracing may be removed until permission is given to Gray from a Registered Professional Engineer.

C. Masonry

A copy of the Bracing Plan needs to be submitted to the Corporate Field Operations team prior to work beginning. The Bracing Plan must include sequencing of erection, material being used as temporary bracing, attachment method, and must be signed by a Principle of the Company. Refer to the bracing requirement set by the Council for Masonry Wall Bracing.

(See Attachment #1 for Bracing Work Procedures)

47. Material Storage

Material shall be stored in an orderly and stable fashion and kept clear of work areas and traffic aisles. Pipe and similar materials shall be stored appropriately as to prevent it from spreading.

Flammable and combustible materials shall be stored at least 25 feet from hot work or other sources of ignition. Fire extinguishers shall be provided at least 25 feet from, but not more than, 75 feet from said storage. Flammable liquids shall be stored in containers marked as such and labeled as to their contents. Flammable liquids, other than in bulk storage, shall be

kept in approved safety cans. Signs indicating “No Smoking within 50 Feet” of flammable material storage shall be posted.

At no time shall propane be stored inside a structure. Also, unlike gasses may not be stored together.

All material stored on the roof shall be stored in a manner that will prevent it from being displaced from movement. Contractor's Site Supervisor /Site Manager shall inspect stored materials on the roof daily and prior to any anticipated wind events.

All joists must be stored properly according to the joist manufactures requirements. Once banding is broken, all of the joist in the bundle must be laid down on their side or secured together to prevent tipping.

48. Material Handling

Loads shall be adequately secured to prevent displacement when being transported by vehicle or equipment. All power industrial truck and forklift use shall follow all requirements set forth by the OSHA 1910.178 Standards. All operators shall meet the training/re- training requirements set forth by the OSHA 1910.178(l)(2) Standards.

All lifting devices must be engineered and labeled with capacities and weight of attachment, if applicable. Custom made lifting devices must be provided with the engineering drawings of the device(s), which shall include the stamp from the Registered Professional Engineer that designed the custom lifting device. Lifting devices must be secured to the equipment to avoid displacement. Tag lines shall be used to control all loads being hoisted overhead. Loads shall not be hoisted or swung over employees' heads.

49. Hand and Arm Protection

Gray requires appropriate gloves to be used while performing any construction activity. Gloves shall be ANSI, Cut-Resistant

Level 4. Subcontractors are required to establish a glove policy that will provide education to their employees in the proper selection, hand placement and use of hand protection. Subcontractor must supply the appropriate gloves as required for the hazards associated for the work. Other types or additional gloves may be required based on the hazard. Other glove types may include, but are not limited to:

- Electrical
- Chemical resistant
- Leather (welding)

In addition to the cut resistant glove requirements, specific activities may require the use of cut resistant sleeves to protect the arms. The Subcontractor must take the initiative in the recognition of when cut resistant arm protection is needed. The Gray Site Management team will assist in determining the necessity for these sleeves.

Activities that require cut resistant arm protection may include, but are not limited to, the handling of sheet metal, plate glass or other sharp materials which have potential to contact the arm above the wrist. If this PPE is deemed necessary, the Subcontractor performing the work shall be responsible for providing appropriate protective sleeves.

Although other tasks may require a glove other than an ANSI Cut-Resistant Level #4, there are ANSI Cut #4 options for most rubber and leather requirements.

50. Hearing Protection

Approved hearing protection shall be worn when employees are continuously exposed to loud noise. As a general rule, if you need to raise your voice to be heard, hearing protection is necessary. No headphones or earbuds should be used as hearing protection.

51. Working in Extreme Heat

In instances where the Heat Index reaches levels in excess of 95° F, the Project Team shall:

- Work with the Subcontractors to develop a break schedule to ensure that employees working in extreme heat/temperatures are given at least a 10-minute break, every 90 minutes.
- Coordinate and modify work schedules with Subcontractors to minimize workers' exposure to working in extreme heat. When working in California, check the Cal/OSHA heat requirements.

52. Mentor Program

Gray has developed a Mentoring Program to help identify employees on our work sites that may be less experienced or new to the construction industry. These mentoring requirements will:

- Foster skill development;
- Help establish a team approach for new employees;
- Help new employees develop additional knowledge of the industry; and

- Help the Subcontractor's management balance experienced team members and inexperience resources on the project.

When mobilizing to the site, Subcontractors shall provide the Gray Site team with a list of any employees that they have identified as a Mentor during the orientation. These Mentors will be identified by a "Mentor" hardhat sticker provided by Gray.

Any employees that have been in the Subcontractor's trade for a period of 1 year or less, and all temporary employees will be identified as a Mentee. These Mentees will be identified by wearing a green DOT class 2 vest. The Gray Site Management team will provide

the vests necessary for the initial roll out of this policy through April 1, 2021. All vests required for the purpose of this policy shall be provided by the employing Subcontractor, at their sole expense for all projects that commence after April 1, 2021.

The Mentees will be required to be identified as a Mentee for a period of at least 1 year after the employee has been working in the Subcontractor's trade. This period may be extended at the sole discretion of the Gray Site team or Subcontractor. Prior to release from the Mentor Program, the Subcontractor shall submit, in writing, to Gray the name of the employee to be released from the program. Upon review, Gray will give final approval of removal from the program. The Subcontractor can present to the Gray Site team the resume of any temporary worker/employee that the Subcontractor believes falls outside of the less than one year of experience definition, that the Subcontractor would prefer not to be identified with a green vest.

All Mentees shall work under the direct supervision of a designated Mentor. It is not a requirement that a Mentee work under the supervision of the same Mentor for the one-year period.

If a Subcontractor has a Mentor Program that meets or exceeds these requirements, they will be allowed to implement their program on site in lieu of this program. However, the employees under the Subcontractor's Mentorship Program will still be identified by the same green vest.

The Gray Construction Management team may require additional Subcontractor team members to be added to the Mentor Program based on observed performance on the project (including, but not limited to, involvement in incidents, common poor safety decisions, etc.). Duration of an employee who is placed in the Mentor Program for this purpose is based on performance and agreed upon by the Gray Management team and the Subcontractor Supervision.

Additionally, there may be circumstances where Gray prohibits Mentees from performing certain, critical, or high-risk tasks which may include, but not be limited to crane operation, live electrical work, or entering a permit required confined space.

Failure of a Subcontractor to identify employees meeting the requirements to be placed in the Mentor Program and/or failure to enforce the program, as directed, will result in disciplinary action of the Subcontractor to include possible removal of supervision from the project.

53. Stop the Drop

When an aerial lift is being used to install material, such as piping, duct work, etc., a rack or support that has been approved for use by the manufacturer shall be used to secure the material in the lift.

If there are no attachments (rack/support) available for the equipment being used for installing/transporting material, the Subcontractor shall devise a safe work plan for installing/transporting their material. The safe work plan shall be presented and reviewed by the Gray Site Team before the work takes place.

When engineering or administrative controls cannot be used to prevent injury due to dropped tools or objects then tool tethering must be used based on the following guidelines.

- Tools 5lbs and under shall be tethered to the person, lift or structure to prevent injury from falling tools.
- Tools 5lbs and over must be tethered to a lift or structure and shall not be tethered to a person's body.
- Acceptable tethers may consist of lanyards, coated wire rope with carabiners, elastic cords, or similar tethers. All tethers must be used in accordance with weight capacities identified by the manufacturer.

When feasible, the aerial lift shall be parked directly under the work in order to prevent objects from falling to the floor below. Spotters, barricades, or tool tethers shall be used for all elevated work that could pose a threat to employees working below. All overhead work shall be barricaded with red danger tape and proper signage, or a spotter(s) must be present to keep people out of the area.

54. Lightning Policy

Lightning should be monitored using some type of mobile app or website (*WeatherBug as an example). When lightning is within 6 miles of the project, all outside, ground activities shall be shut down. When lightning is within 10 miles of the project, all outside, elevated work shall be shut down. This will include, but is not limited to steel erection, roof work, and crane activities. When the lightning is outside the 6-to-10-mile range, these tasks may resume.

55. Pneumatic and Hydrostatic Testing of Piping Systems

This high-risk activity must be evaluated based on code/manufacture requirements with the Site Management team and/or Gray Site Safety Professional to determine specific safety procedures based on pipe size, test medium, volume, area, and site-specific concerns. For all piping systems a written test procedure shall be submitted for the Gray Site Management team to

review. This procedure shall be approved prior to any testing.

If using pressurized air or gas, please reference the "PNEUMATIC or COMPRESSED GAS TEST EXCLUSION ZONE CALCULATION SHEET." This form will calculate the appropriate exclusion zone criteria for the activity. The Gray Site Management team will provide this form.

If doing a hydrostatic test, please reference the "HYDROSTATIC PRESSURE TEST EXCLUSION ZONE CALCULATION SHEET." This form will calculate the appropriate exclusion zone criteria for the activity. The Gray Site Management team will provide this form. On some hydrostatic testing calculations, the barricade area may be exceedingly small. In this instance, establish a barricaded area large enough to safely encompass the area and impact the adjacent work as little as possible.

Once an exclusion zone perimeter distance is established using one of the calculation sheets, proper barricading shall be placed to prevent unauthorized personnel from entering the exclusion zone. The barricade shall be tagged with the Subcontractor's information including the Subcontractor's contact name, contact phone number, and identification of the specific hazard. As a best practice this activity should be considered for off peak hours (nights/weekends) to minimize impact to site.

During the testing phase, if deficiencies are observed on the system being tested, the affected line(s) and/or system shall be put into a zero-energy state, confirmed, and properly locked out before anyone is allowed to enter the area to fix the observed deficiencies.

56. Contractor's Responsibility for Site Safety

We all have a responsibility to ensure project security for our employees, the project site, and our customer's facility. It is important to report all theft or unauthorized personnel or suspicious activity to the Gray Site Management team. All Subcontractors shall give notice to the Gray Site Management team prior to removing or laying-off any employee(s). This notice shall include a list of all employee names that will be affected by the removal or lay off. This is required to ensure no unauthorized personnel come back onto the site and so that the Gray Site Management team is aware of any potential disgruntled employees that may threaten the security of the project. In the case where projects distribute badges or access cards, all badges or access cards must be returned to Gray. If security guards have been employed on site, the names of the dismissed employees will be provided to security so access to the site can be denied, if necessary.

57. Carbon Monoxide

Carbon Monoxide ("CO") is a colorless, odorless, and tasteless hazardous gas that accumulates in enclosed spaces as a result of internal combustion engines. This occurs with the use of gas, diesel, and propane powered equipment. When this equipment is used in an enclosed space or building, a hazardous atmosphere can occur. CO is slightly lighter than air so it will tend to accumulate at higher levels of the room. Workers at elevated heights are at a higher risk of CO poisoning. Several steps can be taken to prevent this from occurring:

- Once the building or area is enclosed and does not have natural ventilation, all gas-powered engines need to be removed and used outside (generators, mixers, gas powered saws, etc.).
- All diesel and propane powered equipment used inside must be low emission (equipped with an exhaust scrubber which works properly).
- Electrical equipment should be used, when available.
- CO monitors should be mounted in suspect areas. OSHA PEL is 50PPM of CO (TWA over 8Hrs) with an action level of 25PPM. If readings are above 25PPM, action must be taken to ventilate the area. These types of monitors are available at local hardware stores (state OSHA plans may be more stringent).
- Try to maintain natural ventilation where available. Leave doors/windows opened when feasible.
- If natural ventilation is not an option and the hazard is still present, mechanical ventilation will be necessary.
- All CO complaints should be taken seriously.

58. Health Crisis

If there is a health crisis that could affect the physical well-being of employees, the Subcontractor will be required to create a plan that must be submitted to Gray for review. A health crisis may include, but is not limited to SARS, COVID, H1N1 or any other illness that may have a significant impact on the project. At a minimum, the plan should include:

- How sick or ill workers will be identified.
- Protocols for the removal and remobilization for sick employees.
- Any safe work plans to minimize the spread of the illness.
- How cleaning and disinfecting of shared tools and equipment will take place and how often.
- Any additional PPE requirements.
- A plan for demobilization should it be required; and
- How contact tracing will be performed.

In addition to these elements, the Subcontractor shall follow all federal, state, and local applicable laws and guidelines. The Subcontractor shall also comply with all of Gray policies as well as all owner requirements.

59. Steel Decking

Wind tacking deck is to secure each sheet at six (6) locations per deck sheet (2 at each end, and 2 in the center).

When winds exceed 20 mph, all decking operations will stop.

Any deck that has been laid will at a minimum will be wind tacked in 6 locations per deck sheet or fully connected.

Any broken bundles will be secured with ratchet straps attached to the steel structure at each end.

Decking being laid out should not exceed the following guidelines without being wind tacked.

- Roof Deck (20 ga or lighter)
 - Winds calm to 10 mph: no more than 3-4 bays
 - Winds 10 to 14 mph: no more than 1-2 bays
 - Winds 14 to 20 mph: no more than 1 bay
- Roof Deck (18 ga or heavier)
 - Winds calm to 14 mph: no more than 3-4 bays
 - Winds 14 to 20 mph: no more than 1-2 bays
- Form Deck (24, 26 and 28 ga)
 - Winds calm to 8 mph: no more than 1-2 bays
 - Winds 8 to 16 mph: no more than 1 bay
- Composite Deck (all gages)
 - Winds calm to 14 mph: no more than 1-2 bays
 - Winds 14 to 20 mph: no more than 1 bay

My Commitment to Safety Acknowledgment

I have thoroughly read all the information in this Subcontractor Safety Orientation Manual provided by Gray. I am committed to following through with all training procedures for my employees. I will become familiar with OSHA regulations and Gray's safety policies outlined in this manual. I realize ignorance of these regulations and policies is no excuse for noncompliance.

I join Gray in the goal of zero incidents for all projects.

I pledge my Company's commitment to providing a safe and clean workplace for all employees on this project.

Project Name

Project Number

Company Name

Gray Construction, Inc.

Subcontractor's Signature

Site Manager/Safety Representative's Signature

Date

Date

This is to be signed by the Foreman, Supervisor, Site Manager, Project Manager, Safety Representative, or any employee in a leadership role during the Site Safety Orientation.

Attachment #1: Work Procedures

1. Precast Tilt-Up Temporary Bracing Work Procedures

Bid Requirements:

- A copy of the temporary bracing requirements will be supplied to all bidders prior to submitting their quotes. This will ensure that proper communication and the requirements are known by all bidders prior to their quoting of the work.

Pre-Award Meeting:

- The Project Manager will communicate all temporary bracing requirements to the Tilt-Up Contractor's Project Manager.

Pre-Construction Meeting:

- Gray's Site Manager will perform a Pre-Construction meeting with the Tilt-Up Subcontractor's Onsite Supervision to communicate the temporary bracing requirements and the means, methods, sequencing, and project requirements. **No lifting of panels will commence until a Pre-Construction meeting has been performed.**

Pre-Lift Meeting:

- The Subcontractor's Competent Person will perform a Pre-Lift meeting with all the Subcontractor's employees. This is to communicate the proper erection and temporary bracing practices and to demonstrate their competency. If new employees are introduced to the project, the Subcontractor is required to train the employees in the Tilt-Up process, including but not limited to the means and method to be used to achieve the proper torque of brace bolts. **No lifting of panels will commence until a Pre-Lift meeting has been performed.**

Temporary Bracing Requirements:

Design:

The Subcontractor is required to provide a full design of the temporary bracing for each individual panel, along with information on all components, such as braces, tilt-wall anchors, brace inserts, and the bolts used for bracing. All components will be designed and provided by one manufacturer to ensure that the components are compatible to one another. A licensed structural engineer shall stamp the full bracing submittal.

The temporary bracing plan shall be designed to withstand wind speeds of 72 mph. If the braces are attached to the floor slab, then:

- The building structural engineer will be required to design floor slabs to support the tilt-up panels for wind loads of 50 mph.
- The building structural engineer will provide acceptable locations for the tilt brace to floor location.

Spread footing elevations shall be in accordance with ACI 117 specifications. Shim packs over 2 inches in height must be of steel construction.

Submittals:

The following information will be submitted to Gray at least 2 weeks prior to the Pre-Construction Tilt-Up meeting.

- The Tilt-Up Subcontractor is to supply a copy of the bracing plan in its entirety, which will include information on all panels, type, and number of braces to be used per panel, panel inserts, bolts for brace to tilt panel connection, and brace-to-ground connection information. It is important to note the torque value of all bracing bolts.
- The name and resume of the Subcontractor's Competent Person.
- Type and size of crane and the annual inspection of that crane.
- The crane operator's certification and years of experience with lifting Tilt-Up panels. Crane operators will be required to have at least 2 years of experience in Tilt-Up.

General Requirements:

- Only new (not previously used) bolts may be used for temporary bracing. The bolts may be re-used only for that specific project. All bolts must be inspected for damage or wear prior to installation.
- Coil bolts shall be installed according to manufacturer's specifications, including torque.

- Gray's Site Manager shall be informed prior to deviating from the plan. This includes the installation of alternate anchorage points for bracing connections. The Site Manager will communicate any deviations to PM, DM, Building Structural Engineer, and the Engineer that approved the bracing submittal.
- The Subcontractor will establish and maintain a restricted access zone until the building structural engineer approves the removal of temporary bracing. The restricted access zone shall be the height of the panel plus 6 feet on the brace and non-brace side. The exclusion zone on the brace side may be moved within 6 feet of the braces but this smaller exclusion zone is prohibited within 100 feet of panels being set on the panel opposite the braces. The restricted access zone shall be marked by, at a minimum, plastic fencing type material of a height of not less than 36 inches. No one is permitted in the restricted access zone without
- first contacting the Gray Site Manager. The area on the brace side shall be marked with danger tape, and work in this area will be allowed if it can be done without disturbing the braces.
- The Subcontractor's Competent Person shall perform and document a daily visual inspection of all braces and anchorages prior to performing any other work.
- The restricted access zone and the building area in proximity to erected panels (height of the panels plus 6 feet in any direction) shall be evacuated if wind speeds reach a peak gust of 35 mph. The wind speed shall be measured at gray's jobsite trailer/office.
- Work will not resume until peak wind speeds have remained below 30 mph for a period of at least 30 minutes.
- If the work area must be evacuated, the Subcontractor's Competent Person will perform a complete "hands-on" inspection of all braces and anchorages prior to returning to work. This is required even if a visual inspection has already been completed that day.

2. Steel Erection Temporary Bracing Work Procedure

Bid Requirements:

- A copy of the temporary bracing requirements will be supplied to all bidders prior to submitting their quotes. This will ensure that proper communication and the requirements are known to all bidders prior to quoting the work.

Pre-Award Meeting:

- The Project Manager will communicate all temporary bracing requirements to the Steel Erection Contractor's Project Manager.

Pre-Construction Meeting:

- Gray's Site Manager will perform a Pre-Construction meeting with the Steel Erection Subcontractor's Onsite Supervision to communicate the temporary bracing requirements and the means, methods, sequencing, and project requirements. **No steel erection will commence until a Pre-construction meeting has been performed.**

Temporary Bracing Requirements:

Design:

The Steel Erection Subcontractor is required to provide a temporary bracing plan to support the building through erection and until the structural engineer provides a release to remove the temporary bracing.

The temporary bracing plan will be created by and installed under the supervision of a Competent Person. The plan must be in writing, contain details on the size of cable, the types of connections to the columns, and the number of clamps or other holding devices. In each case, cable looped around the columns must be installed in such a method to ensure the cable does not slide up or down the column. The plan must take into account the location of the construction, the anticipated conditions (weather) for the area during construction and be able to meet these conditions.

Any deviation of the submitted plan must be communicated to Gray's Site Management team and approved by the Competent Person prior to making any deviations.

Submittals:

The Temporary Bracing Plan must be submitted at least 2 weeks prior to the Pre-Construction meeting. The plan must be signed by an Officer of the company that Gray is contracting with for the steel erection.

General Requirements:

- A Competent Person supplied by the erector, shall be required to inspect, and document a daily inspection on the bracing prior to starting work.
- All areas of the building in proximity of temporary braced steel shall be evacuated when the wind speed reaches 35 mph. Proximity is

defined as the height of the steel structure plus 10 feet as measured in any direction from the base of each erected column.

- The wind speed shall be measured at Gray's job site trailer.
- Work shall not resume until the peak wind gusts have remained below 30 mph for at least 30 minutes.
- If the work area must be evacuated, the erector's Competent Person shall be required to perform a "hands-on" inspection of the temporary bracing prior to returning to work. This is required even if a visual inspection has already been done for the day.
- No column shall be left free standing (overnight, lunch break, or extended duration).
- All temporary cable braces are required to be flagged the full length of the cable.

3. Masonry Temporary Bracing Work Procedure

Bid Requirements:

- A copy of the temporary bracing requirements will be supplied to all bidders prior to submitting their quotes. This will ensure that proper communication and the requirements are known to all bidders prior to quoting the work.
- Pre-Award Meeting:
- The Project Manager will communicate all temporary bracing requirements to the Masonry Contractor's Project Manager.
- Pre-Construction Meeting:
- Gray's Site Manager will perform a Pre-Construction meeting with the Masonry Subcontractor's Onsite Supervision to communicate the temporary bracing requirements and the means, methods, sequencing, and project requirements.
- No masonry work will commence until a Pre-Construction meeting has been performed.

Temporary Bracing Requirements:

Design:

The Bracing Plan can be created using empirical data or as calculated by a structural engineer. Reference manuals for plan development shall be the "Standard Practice for Bracing Walls Under Construction" and "Masonry Wall Bracing Design Handbook", by the Mason Contractors Association of America. If empirical data is used, an Officer of the Subcontractor's company must sign and approve the data. If the information is calculated by a structural engineer, the engineer will stamp the information. Temporary bracing must be determined to resist a wind speed of 40 mph.

Any deviation of the submitted plan must be communicated to Gray's Site Management team and approved by the structural engineer that stamped the original submitted plan.

Submittals:

The Temporary Bracing Plan must be submitted at least 2 weeks prior to the Pre-Construction meeting and must be signed by an Officer of the company that Gray is contracting with for the masonry work.

The Subcontractor will submit information on the temporary bracing components and dead men required for temporary bracing. The Subcontractor is responsible for all components pertaining to temporary bracing.

General Requirements:

- The Subcontractor's Competent Person shall perform a daily inspection prior to performing any other work that day.
- The restricted access zone shall be identified, at a minimum, by danger tape. The tape shall be erected at a height of not less than 39 inches. The restricted zone is defined as the area on each side of a wall subject to the effect of a masonry wall collapse, measured by a horizontal distance equal to the height of the constructed wall plus 4 feet, measured at right angles to the wall, and continuing for the length of the wall. All barriers shall be inspected prior to starting work and any barrier not meeting the requirements of this paragraph shall be repaired or replaced prior to commencing work that day.
- The restricted access zone shall be evacuated when the wind speed reaches 35 mph.
- All wind speeds shall be measured at Gray's jobsite trailer.
- Work shall not resume until the measured wind gust speed has remained below 30 mph for at least 30 minutes.
- The Competent Person shall perform an inspection after each wind event that resulted in the evacuation of the restricted area. This is to be done even if a daily inspection had already been done that day.

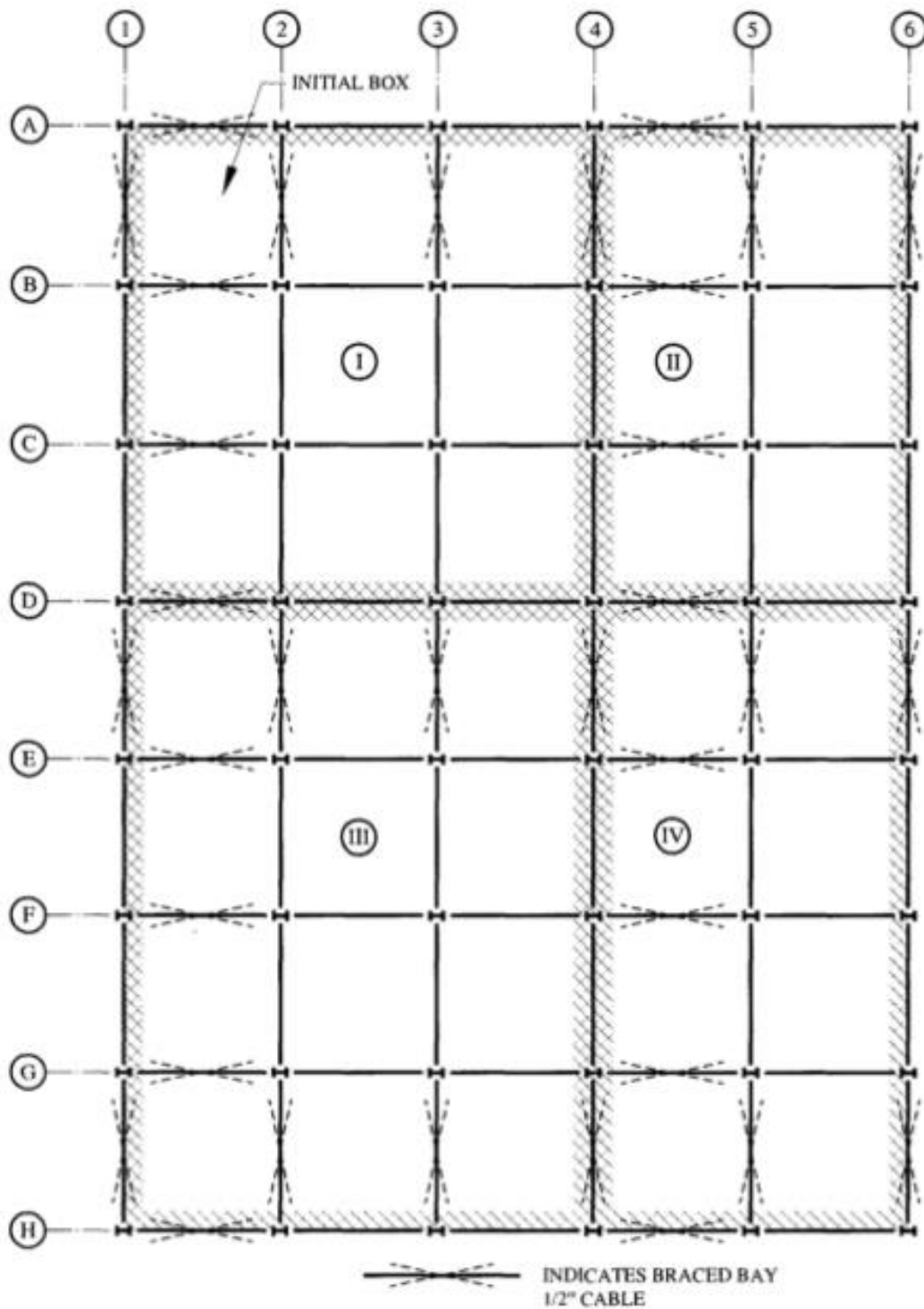


Fig. 7.1 Erection Plan

AISC Design Guide 10 Erection Bracing of Low-Rise Structural Steel Buildings

Attachment #2: Gray's Drug-Free Workplace Summary

1. Prohibited Conduct:

To the extent permitted by federal, state, and local laws, the following behaviors are prohibited on a Gray job site:

- Use, possession, manufacture, distribution, attempted distribution, dispensation, attempted dispensation, sale, attempted sale, purchase, attempted purchase, cultivation, or storage, or being "under the influence" of illicit drugs (defined as a positive test result);
- Unauthorized use, possession, or being "under the influence" of alcohol (defined as a breath alcohol concentration of .02 or higher, unless otherwise specified in the Permitted Conduct section of this policy summary);
- Conviction for any criminal drug or alcohol statute for a violation occurring in the workplace, while conducting company business, while driving company owned, rented, or leased vehicles or personal vehicles being used for company business;
- Failing to notify a Vice President/Manager or Human Resources of any criminal, drug or alcohol statute conviction or arrest within 24 hours or the next working day for a violation occurring in the workplace, while conducting company business or while driving for company business;
- Failing to report any change in driver's license status, within 24 hours or the next working day, to his/her supervisor, if his/her job function may include driving a vehicle for company business;
- Taking a prescription drug that is not according to their physician's direction;
- Refusing to consent to, remain ready for, cooperate with, submit to, or tampering with a drug and/or alcohol specimen or testing;
- Having any drug or alcohol statute conviction or arrest or engaging in the following conduct, either off company premises or during off-duty hours:
 - Possession, use, manufacture, distribution, dispensation, cultivation, or sale of controlled substances, illegally used drugs, or alcohol off company premises that may adversely affect the company, the team member's work performance, or the team member's safety, others' safety at work or the general public;
 - Illegal use of legal substances off company premises or during off-duty hours that may adversely affect the company, the team member's work performance, the team member's safety, or others' safety at work.

2. Kinds of Testing

The only way to know with certainty if an individual is under the influence of drugs and/or alcohol is to conduct a test. The primary method used to determine the presence of drugs in the system under this policy includes a urine test; the presence of alcohol is determined by a breath alcohol test. For the safety of all our team members, and in compliance with all federal, state, and local laws, Gray may test for drugs and/or alcohol in the following circumstances: pre-employment, during the probationary period, reasonable suspicion, post-accident, random, return from layoff, as required by the customer or government, and as a follow-up to treatment or assessment.

Gray requires a DOT 11-Panel Extended Opiates Drug Screen administered in compliance with SAMHSA (Substance Abuse & Mental Health Service Administration) or NIDA (National Institute on Drug Abuse) requirements. The 11-Panel Extended Opiates Drug Screen includes Amphetamines; Barbiturates; Benzodiazepines; Buprenorphine / Metabolite; Cannabinoids; Cocaine / Metabolites; MDMA; Methadone / Metabolite; Opiates; Oxycodone / Oxymorphone; Phencyclidine.

For Breath Alcohol testing, if sending for reasonable suspicion, the supervisors that are sending team member, should specify what their reasonable suspicion covers. If for post-accident, should mark that testing needs to be for both.

3. Consequences

Any violation of Gray's DFWPP, even a first offense, may be a basis for disciplinary action, up to and including termination. However, particularly serious violations, such as selling drugs on company property while conducting business for Gray or on our jobsites will normally result in immediate termination and referral for criminal prosecution.

4. Inspections

Should Gray have reason to believe that an individual may be in possession of alcohol, drugs, or drug paraphernalia on company property or on company time, Gray may search company property or personal effects or vehicle while on company property or on company time.

5. Safety-Sensitive Positions

Any position, which by the nature of the work involved is accompanied by such risk, that even a momentary lapse of attention could have profound consequences to the safety of the co-worker, the other co-workers, the customers, the company, or the general public. For purposes of this policy, the following are the considered safety-sensitive positions:

- All personnel driving a company or personal vehicle for work regardless of whether it is a rented or leased vehicle
- All personnel on jobsites
- All personnel assigned to the service team or maintenance personnel
- All self-perform team members

6. Confidentiality

All information concerning drug or alcohol testing referrals and testing results, or treatment and rehabilitation of a team member will be kept confidential.

7. Reservation of Rights

Gray reserves the right to interpret, change, or rescind the policy in whole or in part, with or without notice. In addition, changes to applicable federal or state laws or regulations may require Gray to modify or supplement the policy. For more information on this policy, you may request a copy of Gray's DFWPP.

8. Random Testing

In Gray's sole, but reasonable discretion, Gray may require Subcontractor (at Subcontractor's sole cost) to implement pre-access, random, reasonable suspicion, or post incident drug testing that (1) randomly tests at least 5% of the Subcontractor's employees or one of the Subcontractor's tiered contractor's employees at least once a month; and (2) submit a letter from the Subcontractor's HRdepartment confirming compliance with this Gray's DFWPP.

Attachment # 3: Americans with Disabilities Act

The ADA protects the rights of the disabled and mandates that disabled individuals have equal access to employment opportunities. Consistent with the ADA, Gray is an equal opportunity employer that requires all contractors and employees to adhere to a discrimination and harassment free workplace, to protect the privacy of disabled workers, and to provide reasonable accommodations to qualified individuals with disabilities to ensure that they can perform the essential functions of the job without creating an undue safety risk of accident or injury on construction projects and job sites. By following the recommendations of the ADA, as well as recommended safety protections established by national safety organizations, Gray and its contractors can protect the rights of all workers, including disabled workers, while mitigating the risk of an accident or injury at a construction site.

To do so, as an employer you must not make any assumptions about an employee's abilities or medical conditions. Instead, you should:

- Consider the project as a whole and outline the essential functions, including physical requirements, of the anticipated jobs to be performed on the project site without regard to any specific employee's or individual's condition or characteristics. Essential functions are defined as the fundamental job duties of the employment position. A job function is essential if the job exists to perform that function. The ADA permits an employer to establish job related qualification standards, including the physical and mental standards necessary for job performance, health, and safety. Those standards, however, must not screen out, or tend to
- screen out, individuals on the basis of disability unless they are job-related and consistent with business necessity. For example, you can require that employees meet national industry safety guidelines. But you cannot have a rule that bans all people with hearing impairments from a job as such a rule is based solely on a disability and on the misconception that hearing deficiencies cannot be sufficiently corrected by hearing aids.
- Identify the safety risks of accident and injury created by the project and/or the specific anticipated jobs to be performed on the project and put into place appropriate control measures to mitigate these risks using industry or scientific objective data.
- Designate specific safety and/or human resources personnel to observe workers on the project and to review fitness for duty exams and other medical information concerning employees working on the site and periodically assess whether all the safety risks to all employees, including those with medical conditions or other limitations that create a risk of injury or accident, have been assessed properly. If not, implement appropriate control measures for the entire project or group of workers performing the same type of jobs using industry or other scientific objective data.
- If an individual's physical limitations, disability, or medical condition otherwise prevents him/her from performing the essential functions of the job and adhering to the identified control risk measures, conduct an individualized inquiry with the employee (in conjunction with HR and/or legal counsel) to determine if the employee can perform the essential functions of the job with a reasonable accommodation that does not create a direct threat of harm. You should NOT make any employment decisions
- without conducting this individualized assessment, nor should you apply a blanket rule or prohibition based upon an individual's age, ability to hear, see, or walk, that otherwise adversely impacts that individuals' employment as same runs afoul with the ADA.
- An individual with a disability is not qualified for a specific employment position if he or she poses a "direct threat" to his or her own health and safety or to others. Specifically, a direct threat is a situation presenting a significant risk of substantial harm
- to the health or safety of the employee or others that cannot be eliminated or reduced by a reasonable accommodation. An elevated risk of injury is not sufficient, and the risk must be current, not speculative, or remote. The assessment must be based on objective, scientific evidence, not on subjective perceptions, irrational fears, or stereotypes. The determination must be made on a case-by-case basis and relevant evidence may include input from the employee, medical doctors, rehabilitation counselors, or physical therapists with expertise in the disability or direct knowledge of the individual with a disability. Be sure to consult with the employee themselves and colleagues, seeking opinions and ensuring they are involved in discussions which affect them. Those involved in the work often propose good solutions. Proper factors to consider when conducting the individualized direct threat analysis include:
 - Does the employee's condition create a direct threat to their health and safety or the health and safety of others?
 - What is the duration of the risk based on medical data?
 - What is the nature and severity of the potential harm based on industry and national safety standards?
 - If the condition is professionally managed with medication or other safety protocols, the employer may conclude that review of the situation through periodic medical updates at regular intervals is sufficient.

- What is the likelihood that the potential harm will occur?
- Can these risks be prevented or adequately controlled through normal health and safety management?
- Can the risks be addressed by allowing other colleagues to do certain elements of the activity, or by providing suitable, alternative equipment, for example, automated equipment to reduce manual handling or change systems of work?
- Is the potential harm imminent?
- If not, what reasonable adjustments could be put in place to prevent or adequately control the residual risks?

Staying alert and proactive in identifying and addressing workplace risks and concerns can help reduce safety accidents and injuries on a job site. However, we must make sure we adhere to the rights of all employees and workers when addressing these safety risks and concerns. Relying on industry standards, the most current medical knowledge, and the best available objective when conducting individualized assessments of an individual's ability to perform the essential functions of a particular job on a job site will ensure compliance with the law and help us mitigate these safety risks and concerns.